

HTT Mutation Detection & Gene Regulatory Network

A Computational Framework for Huntington's Disease Analysis

Computational Biology Research Team

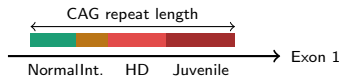
Based on CFG/PCFG Parsing & ODE Simulation

May 9, 2026

References: Langbehn (2004), Goold et al. (2021), Wang et al. (2024),
Zuccato et al. (2003), Steffan et al. (2000)

- 1 Biological Background
- 2 System Architecture
- 3 CFG/PCFG Algorithms
- 4 Fitness Landscape
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- 6 Protein Structure Prediction
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- **HTT gene** on chromosome 4p16.3 encodes huntingtin protein
- **CAG trinucleotide repeat** in exon 1 → polyglutamine (polyQ) tract
- **Thresholds** (Langbehn et al., 2004; Semaka et al., 2016):
 - Normal: ≤ 35 CAG
 - Intermediate: 36–39 CAG (reduced penetrance)
 - Full penetrance: 40–59 CAG
 - Juvenile: ≥ 60 CAG
- **Anticipation:** Earlier onset in successive generations
- Paternal transmission shows larger expansions (+2.5 CAG/gen)



Key references:

- Langbehn DR et al., *Clin Genet* 2004;65:267–277
- Bates GP et al., *Nat Rev Dis Primers* 2015;1:15005

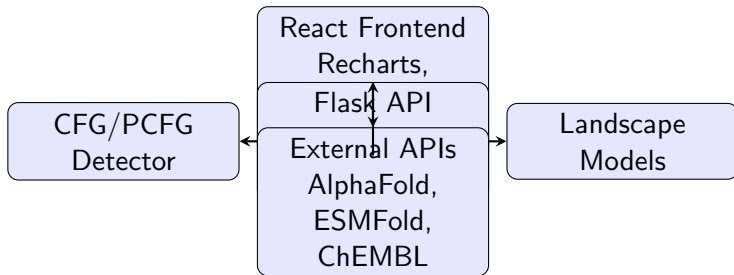
Toxic gain-of-function mechanisms:

- ① **Protein aggregation** — polyQ expanded tracts form β -sheet-rich aggregates (Scherzinger et al., 1997)
- ② **Transcriptional dysregulation** — sequestration of:
 - CBP/p300 (Steffan et al., 2000)
 - Sp1 (Buckley et al., 2010)
 - TBP/TAFII130 (Nucifora et al., 2001)
- ③ **Mitochondrial dysfunction** — DRP1 impairment, reduced PGC-1 α (Cui et al., 2006)
- ④ **Proteostasis collapse** — UPS overload, autophagy impairment

Loss-of-function:

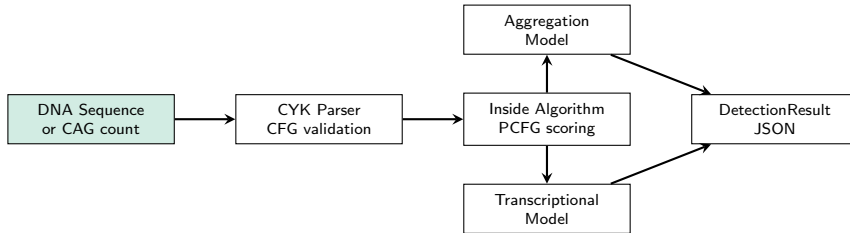
- wtHTT sequesters REST in cytoplasm $\rightarrow$ absent REST $\rightarrow$ $\uparrow$ BDNF
- mHTT cannot sequester REST $\rightarrow$ nuclear REST $\rightarrow$ $\downarrow$ BDNF (Zuccato et al., 2003)
- $\sim$ 2000 neuronal genes silenced via NRSE elements

Reference: Zuccato C & Cattaneo E, *Nat Rev Neurosci* 2007;8:801–812



Key technologies:

- Frontend: React 18, Recharts (D3-based), 3Dmol.js, React Three Fiber
- Backend: Python Flask, CORS-enabled, Blueprint architecture
- External: AlphaFold DB, ESMFold (Meta), ChEMBL REST API



Output fields:

- `cag_count`, `ccg_count` — repeat counts
- `normalized_score` ($0 \rightarrow 1$) — grammar fitness
- `anomaly_score` — $1 - \text{normalized_score}$
- `parse_tree` — hierarchical CFG structure (d3.js compatible)
- `onset_prediction` — Langbehn model estimate

Why CFG? Regular expressions (finite automata) cannot enforce flanking context while counting repeats. CFG can.

Grammar rules (CNF-like):

$S \rightarrow \text{EXON1}$

$\text{EXON1} \rightarrow \text{FLANK5 REPEAT_REGION}$

$\text{REPEAT_REGION} \rightarrow \text{CAG_BLOCK FLANK3}$

$\text{CAG_BLOCK} \rightarrow \text{CAG_UNIT} \mid \text{CAG_UNIT CAG_BLOCK}$

$\text{CAG_UNIT} \rightarrow \text{'CAG'}$

$\text{FLANK5} \rightarrow 5' \text{ invariant pattern}$

$\text{FLANK3} \rightarrow 3' \text{ invariant pattern}$

Recursive rule:

$\text{CAG_BLOCK} \rightarrow \text{CAG_UNIT CAG_BLOCK}$

This is what makes the grammar **context-sensitive** — it can count arbitrarily many repeats while maintaining structural context.

Reference: CFG design inspired by RNA secondary structure prediction (Durbin 1998, *Biological Sequence Analysis*)

Cocke-Younger-Kasami (CYK) parser — $O(n^3 \cdot |G|)$ time complexity.

CYK Dynamic Programming Recurrence

For a grammar in Chomsky Normal Form (CNF):

$$\text{dp}[i][j][A] = \bigvee_{k=i}^{j-1} \bigvee_{B \rightarrow C \in \text{rules}} \text{dp}[i][k][B] \wedge \text{dp}[k+1][j][C]$$

where $A \rightarrow BC$ is a production, and terminals handled at $j = i$.

In our implementation:

- Grammar converted to CNF at runtime
- Terminal matching: codons ('CAG', 'CCG') and flank patterns
- Parse success = $\text{dp}[0][n][S] > 0$
- Parse tree reconstructed via backpointers

Reference: Younger DH (1967). Recognition and parsing of context-free languages in time

Probabilistic CFG (PCFG): each rule has a probability $\theta_{A \rightarrow \alpha}$

Inside Probability

$$\beta(i, j, A) = P(\text{terminal string } w_i \dots w_j \mid A)$$

Recursive computation:

$$\beta(i, j, A) = \sum_{k=i}^{j-1} \sum_{B, C} \theta_{A \rightarrow BC} \cdot \beta(i, k, B) \cdot \beta(k+1, j, C)$$

Base case: $\beta(i, i, A) = \theta_{A \rightarrow w_i}$

Our PCFG scoring components:

- 1 $\log P(\text{CAG count})$ — from empirical healthy distribution
- 2 $\log P(\text{flanks intact})$ — penalty if pattern missing
- 3 $\log P(\text{no interruptions})$ — each non-CAG codon penalized

Empirical distribution from Metzger et al. (meta-analysis of healthy individuals):

CAG repeats	Frequency	$\log P$ (natural)
6–10	0.001–0.015	–6.9 to –4.2
11–16	0.020–0.082	–3.9 to –2.5
17–19	0.095–0.092	–2.35 to –2.38
20–26	0.085–0.017	–2.46 to –4.07
27–35	0.010–0.0003	–4.6 to –8.1
≥ 36	$\sim 10^{-8}$ (extrapolated)	< -18.4

Peak at 17–19 repeats — corresponds to 18 ± 2 polyQ in healthy huntingtin.

Reference: Metzger S et al. (2016). The distribution of CAG repeats in the general population. *Eur J Hum Genet*.

Definition: Cellular fitness = probability that a cell with given mHTT CAG length survives and functions normally.

Weighted Composition

$$\text{Fitness}(n) = w_1 \cdot \text{PCFG}(n) + w_2 \cdot (1 - \text{Agg}(n)) + w_3 \cdot (1 - \text{Dys}(n))$$

with $w_1 = 0.30$, $w_2 = 0.40$, $w_3 = 0.30$.

PCFG fitness:

$$\frac{P_{\text{healthy}}(n)}{\max P}$$

Grammar-based health proxy.

Aggregation propensity:

$$\text{Agg}(n) = 1 - e^{-k(n-35)} \quad (n > 35)$$

$k = 0.15$ (Bhattacharyya, 2005)

Transcriptional dysregulation:

$$\text{Dys}(n) = 1 - e^{-0.08(n-40)} \quad (n > 40)$$

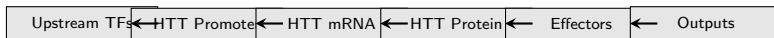
CBP/p300 sequestration effect.

[Screenshot of LineChart from LandscapeCard.jsx]

Graph elements explained:

- **X-axis:** CAG repeat length (1 to 80+)
- **Y-axis:** Normalized fitness (0 = lethal, 1 = healthy)
- **Blue line:** Composite fitness (monotonic decreasing)
- **Orange/red/purple reference lines:** Clinical thresholds at CAG 35, 40, 60
- **Red dot:** User's query CAG (from detection)
- **Tooltip:** Shows exact fitness value on hover

71 nodes, 92 edges — ODE simulation of disease vs. healthy states



Layers:

- 1 **Upstream regulators:** Sp1, CTCF, NF- κ B, STAT1 (bind HTT promoter)
- 2 **HTT promoter:** TATA-less, GC-rich, 20 bp tandem repeats
- 3 **mRNA:** HTT transcript + antisense (HTT-AS)
- 4 **Protein:** wtHTT ($\leq 35Q$) vs mHTT ($\geq 40Q$)
- 5 **Effectors:** REST, CBP, p300, p53, BDNF, PGC-1 α , SIRT1, HSF1
- 6 **Outputs:** Mitochondrial genes, apoptosis, MSN death

Generalized ODE for each node:

$$\frac{dx_i}{dt} = \sum_{\text{inputs}} \gamma_j \cdot h_{\text{type}}(x_j, K_j, n_j) - \delta_i x_i$$

Hill functions:

$$h_{\text{act}}(x, K, n) = \frac{x^n}{K^n + x^n} \quad h_{\text{rep}}(x, K, n) = \frac{K^n}{K^n + x^n}$$

Example: HTT Promoter regulation

$$\frac{d[\text{HTT_PROM}]}{dt} = \underbrace{8 \cdot \text{act}(\text{Sp1}) \cdot \text{act}(\text{CTCF}) \cdot \text{rep}(\text{NF-}\kappa\text{B}) \cdot \text{rep}(\text{STAT1})}_{\text{AND-like integration}} - \delta \cdot [\text{HTT_PROM}]$$

Key biological parameters:

- $K_{\text{Sp1}} = 3.0$, $n = 2$ (cooperative binding to GC-rich promoter)
- $\delta = 0.35$ (protein/mRNA degradation rate)
- Time step $\Delta t = 0.04$ (simulated hours)

NOT gate examples:

- $[REST] \propto h_{rep}([wtHTT])$ — wtHTT sequesters REST
- $[BDNF] \propto h_{rep}([REST])$ — REST represses BDNF

AND gate:

$$HTT_PROM \propto Sp1 \wedge CTCF$$

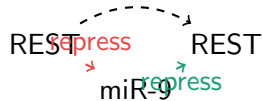
NOR gate:

$$CBP_{free} \propto h_{rep}([mHTT])$$

mHTT sequesters CBP, p300, PCAF simultaneously.

Reference: Gould R et al. (2021). *Cell Reports* — FAN1/MSH3 competitive binding model.

Feed-Forward Loop (FFL):



REST represses miR-9; miR-9 targets REST mRNA → double-negative feedback → bistability.

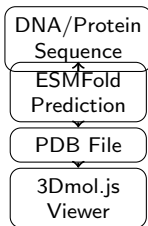
[Screenshot from GRNSimulator.jsx]

Node styles encode function:

- **Diamonds** — Transcription factors
- **Circles** — Proteins (wtHTT/mHTT)
- **Hexagons** — Coactivators/repressors
- **Rectangles** — Pathology outputs

Edge types:

- **Solid** — activation ($A \rightarrow B$)
- **Dashed** — repression ($A \dashv B$)



ESMFold (Meta AI):

- 15 layers, 512M parameters
- $\sim 3\times$ faster than AlphaFold2
- pLDDT confidence per residue

AlphaFold DB: fallback for normal HTT (P42858)

3Dmol.js styles:

- **Cartoon** — α -helix, β -sheet
- **Surface** — SASA
- **Stick** — atomic detail
- pLDDT coloring: blue $\rightarrow$ red

pLDDT (predicted Local Distance Difference Test) — per-residue confidence metric (0–100):

pLDDT	Interpretation	Color
90–100	Very high (correct backbone)	Dark blue
70–90	Confident (good)	Light blue
50–70	Low (may be disordered)	Yellow
< 50	Very low (disordered)	Red

Biological insight: PolyQ regions in HTT are intrinsically disordered ($\text{pLDDT} < 50$) — exactly why they are drug-accessible.

Accessibility score for drug binding:

$$\text{access}(i) = \begin{cases} 1.00 & \text{pLDDT} < 50 \quad (\text{disordered}) \\ 0.60 & 50 \leq \text{pLDDT} < 70 \\ 0.20 & 70 \leq \text{pLDDT} < 90 \\ 0.05 & \text{pLDDT} \geq 90 \end{cases}$$

Model for drug-polyQ binding:

Ligand-receptor binding equilibrium

$$\text{Fraction bound} = \frac{[D]}{[D] + K_d}$$

where K_d is the dissociation constant (concentration at 50% occupancy).

CAG-dependent K_d scaling:

$$K_d(n) = \begin{cases} 200 \mu\text{M} & n \leq 35 \quad (\text{very weak binding}) \\ 50 \cdot e^{-0.08(n-35)} \mu\text{M} & n > 35 \end{cases}$$

- $K_d(40) \approx 50 \mu\text{M}$, $K_d(60) \approx 11 \mu\text{M}$
- Expanded polyQ exposes more hydrophobic surface $\rightarrow$ higher affinity

Dose-response curve (output visualization):

- X-axis: Drug concentration (0–200 μM)
- Y-axis: Bound fraction (0–100%)

Aggregation kinetics model for polyQ:

Simplified Oosawa model

$$\frac{dM}{dt} = k_n \cdot (1 - M)^2 + k_e \cdot M \cdot (1 - M) - k_{\text{off}} \cdot M$$

where:

- M = fraction of aggregated protein
- k_n = nucleation rate (slow, concentration-dependent)
- k_e = elongation rate (faster once nucleus forms)
- k_{off} = dissociation rate

CAG dependence:

$$k_n \propto \text{Agg}(n), \quad k_e \propto \text{Agg}(n)$$

with $\text{Agg}(n) = 1 - e^{-0.15(n-35)}$ for $n > 35$.

Drug effect:

[AreaChart from DrugInteractionPage.jsx]

Graph interpretation:

- **X-axis:** Time (hours) **Y-axis:** Fraction aggregated ($0 \rightarrow 1$)
- **Red curve:** No drug (baseline aggregation)
- **Green curve:** With drug (extended lag phase)
- **Sigmoidal shape** — cooperative nucleation followed by rapid elongation
- **Lag phase** defined as time to 10% aggregation

Biological meaning: The longer the lag phase, the more time cells have to clear misfolded monomers before toxic oligomers form.

Median onset age as function of CAG:

$$\text{Onset}_{\text{median}} = 21.54 + e^{(9.556 - 0.1460 \cdot \text{CAG})} \text{ years}$$

CAG repeats	Median onset (years)
40	60
42	52
44	46
46	40
48	36
50	33
55	25
60	10

Note: CAG length explains ~50–60% of onset variability. Modifier genes (FAN1, MSH3, DNA repair pathways) account for the rest.

Reference: Langbehn DR et al. (2004). *Clin Genet* 65, 267–277.

Paternal transmission:

- Mean expansion: +2.5 CAG
- Standard deviation: 3.5 CAG
- Reason: Spermatogenesis involves many cell divisions
- Mutation frequency: $\sim 70\text{--}82\%$ for disease alleles

Simulation approach:

$$\Delta\text{CAG} \sim \mathcal{N}(\mu_{\text{parent}}, \sigma_{\text{parent}}^2)$$

$$\text{CAG}_{\text{child}} = \lfloor \text{CAG}_{\text{parent}} + \Delta\text{CAG} + 0.5 \rfloor$$

clipped to $[6, 120]$.

Reference: Duyao M et al. (1993). *Nat Genet* 4, 387–392.

Maternal transmission:

- Mean expansion: +0.5 CAG
- Standard deviation: 1.5 CAG
- Oogenesis: fewer divisions, more stable
- Rare contractions possible

API	Usage
AlphaFold DB	Normal HTTP structure (AF-P42858-F1-model_v4.pdb)
ESMFold (Meta)	Mutant protein folding via POST to api.esmatlas.com
ChEMBL	Drug-HTTP interaction queries (target CHEMBL2093872)

Flask proxy pattern:

```
@http_bp.route('/esmfold/fold', methods=['POST'])
def fold_esmfold():
    seq = request.json.get('sequence')
    resp = requests.post(
        'https://api.esmatlas.com/foldSequence/v1/pdb/',
        data=seq, headers={'Content-Type': 'text/plain'},
        timeout=60
    )
```


Component	Algorithm/Model
Mutation detection	CYK parser for CFG, Inside algorithm for PCFG
Fitness landscape	Weighted sum: PCFG + Aggregation + Transcription
GRN simulation	Hill kinetics ODE (71 nodes, 92 edges)
Protein folding	ESMFold (Transformer), pLDDT confidence
Drug binding	Langmuir isotherm, Oosawa nucleation-elongation
Heredity	Autosomal dominant + meiotic instability (Normal)
Onset prediction	Langbehn 2004 survival model

Key biological insights implemented:

- 1 CAG > 35 → protein aggregation + transcriptional dysregulation

- **Graph Neural Networks** for GRN inference from single-cell data
- **Reinforcement Learning** for optimal drug dosing schedules
- **Docking simulations** (AutoDock Vina) for specific compounds
- **Patient-derived organoid data** integration
- **Longitudinal clinical data** for onset model calibration
- **CRISPR editing strategies** to reduce CAG length in vivo

Thank you

Code repository: <https://github.com/username/htt-explorer>

-  Langbehn DR, Brinkman RR, Falush D, Paulsen JS, Hayden MR (2004). A new model for prediction of the age of onset and penetrance for Huntington's disease based on CAG length. *Clin Genet* 65:267–277.
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